Probability and Statistics 1

Tutorial Sheet — Session 1: Solutions

Basic statistical notions and vocabulary; frequency tables and graphs for a qualitative character

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Exercise 1

(a) Population: the 5000 students of the university. Individual: one student. Sample: the 200 students who received the questionnaire. This is a *poll*, since the sample (200) is a strict subset of the population (5000).

(b) Population: the 5000 login attempts of the day. Individual: one login attempt. Sample: all 5000 attempts. This is a *census*, since the sample coincides with the population.

(c) Population: all users of the streaming platform. Individual: one user. Sample: the 1000 selected users. This is a *poll*, since 1000 is a strict subset of several million.

Exercise 2

(a) Qualitative nominal. Modalities: Python, C++, Rust, JavaScript, etc. No modality is "greater" than another.

(b) Quantitative discrete. Possible values: $0, 1, 2, \ldots$ (isolated integers).

(c) Qualitative ordinal. Modalities: low, medium, high, critical. They are ordered, but the differences between consecutive modalities have no numerical meaning.

(d) Quantitative continuous. Possible values: any positive real number, in ms.

(e) Quantitative discrete. Possible values: $1, 2, 4, 8, 16, \ldots$ (isolated integers).

Exercise 3

(a) The completed table is:

Cloud provider	n_i	$f_i = n_i/120$	$p_i = 100\, f_i$
AWS	48	$48/120 = 0.40$	40
Azure	36	$36/120 = 0.30$	30
GCP	24	$24/120 = 0.20$	20
Other	12	$12/120 = 0.10$	10
Total	120	1.00	100

(b) Relative frequencies: $0.40 + 0.30 + 0.20 + 0.10 = 1.00$. Percentages: $40 + 30 + 20 + 10 = 100$. Both checks hold.

(c) The mode is **AWS**, because its absolute frequency (48) is the largest of the four modalities.

Exercise 4

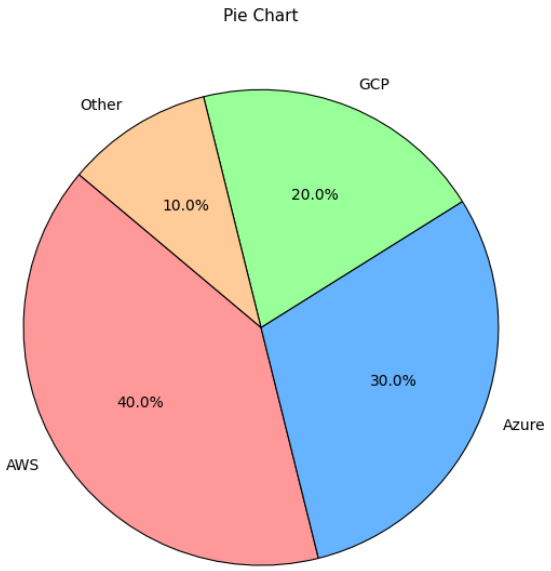
(a) Using $\theta_i = (n_i/120) \times 360°$:

$$\theta_{\text{AWS}} = 0.40 \times 360° = 144°, \qquad \theta_{\text{Azure}} = 0.30 \times 360° = 108°,$$

$$\theta_{\text{GCP}} = 0.20 \times 360° = 72°, \qquad \theta_{\text{Other}} = 0.10 \times 360° = 36°.$$

(b) $144° + 108° + 72° + 36° = 360°$.

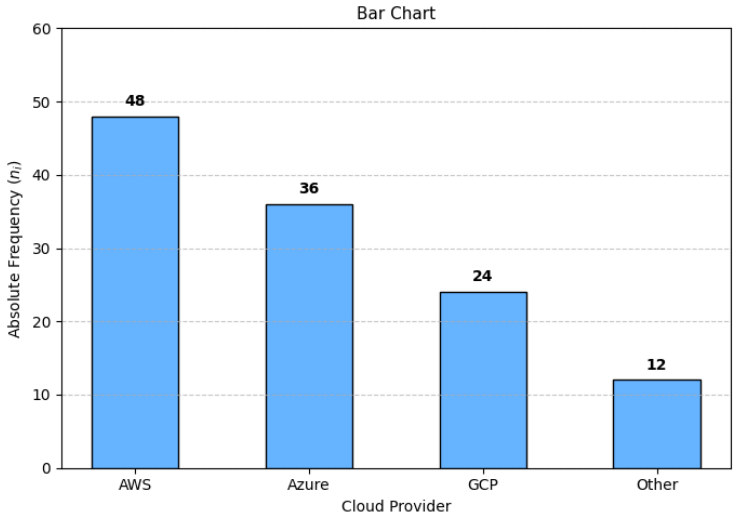
(c) The pie chart has four sectors of 144°, 108°, 72° and 36°. **AWS** occupies the largest sector (144°); **Other** occupies the smallest (36°).



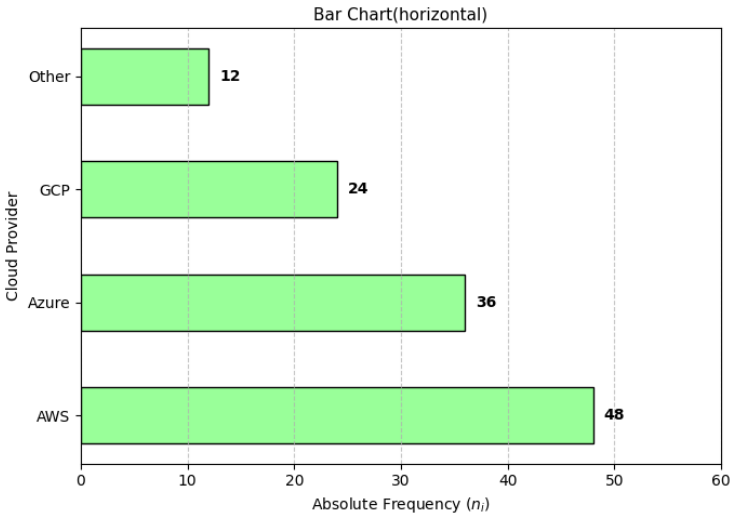
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Exercise 5

(a) The horizontal axis shows the cloud providers (AWS, Azure, GCP, Other) and the vertical axis shows the absolute frequencies (or relative frequencies, or percentages). The vertical axis must start at 0, so that the ratio of two bar heights equals the ratio of the corresponding frequencies. The bars have heights $48, 36, 24, 12$.



(b) In a band diagram, the bars are horizontal: the modalities are listed vertically and the bar lengths are proportional to the frequencies. The bar lengths are again $48, 36, 24, 12$ (or the corresponding f_i or p_i), starting from 0.



(c) A **bar chart** is more appropriate to compare Azure and GCP: the two bars are placed side by side on a common baseline and their heights can be compared directly. On a pie chart, the comparison of two adjacent sectors of 108° and 72° is possible but much less precise, because the eye is not sensitive to small differences of angles. The pie chart is a communication tool; the bar chart is an analysis tool.

Exercise 6

(a) From $n = 200$:

- VS Code: $n_i = 95$, $f_i = 95/200 = 0.475$, $p_i = 47.5$.

- Jupyter Notebook: $f_i = 0.225$, so $n_i = 0.225 \times 200 = 45$, $p_i = 22.5$.
- Vim/Neovim: $p_i = 17.5$, so $f_i = 0.175$ and $n_i = 0.175 \times 200 = 35$.
- emacs: $n_i = 25$, $f_i = 25/200 = 0.125$, $p_i = 12.5$.

The completed table is:

Editor	n_i	$f_i = n_i/200$	$p_i = 100\, f_i$
VS Code	95	0.475	47.5
Jupyter Notebook	45	0.225	22.5
Vim/Neovim	35	0.175	17.5
emacs	25	0.125	12.5
Total	200	1.000	100.0

(b) $95 + 45 + 35 + 25 = 200$. Also $0.475 + 0.225 + 0.175 + 0.125 = 1.000$.

(c) The mode is **VS Code**, with absolute frequency $n_i = 95$, the largest of the four modalities.